BUIDL CTC 2026 FALL · DEFI

Kitty

Savings circles where every payment is proven, not promised.

Ethereum stablecoins · settled and credit-scored on Creditcoin · via the Attestcoin Protocol

PROBLEM

Hundreds of millions save in circles nobody can see.

- Chit funds, susu, tandas, chamas, stokvels: ten friends, one pot a month.
- They fail two ways: the treasurer runs, or a member stops paying and nobody outside the group ever knows.
- Ten years of perfect payments builds zero formal credit history.

TRUST TODAY

Who you have to trust

STEP	INFORMAL CIRCLE	“ROSCA ON-CHAIN” APPS	KITTY
Holds the money	Treasurer	One contract on one chain	Escrow vault on Ethereum
Knows who paid	Treasurer’s notebook	Same contract	Attestcoin-proven txs on Creditcoin
Decides the deadline	Treasurer	block.timestamp / admin	Attested source-chain block height
Picks the recipient	Treasurer / lottery	VRF oracle	Fixed order or by proven score
Credit history	None	None	Kitty Score, readable by any lender

SOLUTION

Money on Ethereum. Rules on Creditcoin. Proofs in between.

- Members pay into a minimal escrow vault on Sepolia.
- The attestor network attests the block; one batch proof covers the whole round.
- KittyLedger verifies through precompile 0x0FD2, decodes and binds every payment, enforces deadlines from 0x0FD3.
- Payouts are proven back. “Paid” is never an operator’s word.
- Any member can prove a round from the browser: the Proof Builder serves CORS, the ledger doesn’t care who submits.

LIVE LOOP

Pay → attest → prove → verify → close → pay out → prove back

Demo: three members, 100 tUSD, round 0 everyone pays; round 1 one member misses; the deadline block is attested; the round closes with a missed record; the score drops from 515 to 395.

ATTESTCOIN DEPTH

What the ledger actually checks

- Batch `verifyAndEmit`: up to 10 txs, one continuity proof, one call for up to ten payments pooled across every open circle, preflighted free with the view verify, 24% cheaper than singles on the live 0x0FD2.
- EvmV1Decoder: receipt status 1, exactly one Contributed log, emitter = registered vault, tx.to = vault, tx.from = member, exact amount, current round.
- Query id = keccak(chainKey || height || txIndex), same as ASCBase; per-circle chain key validated with `get_chain_by_key`; vault allowlist keyed by chain.
- Deadlines from attested height: `is_height_attested(chainKey, deadline + 64)`.
- The dashboard asks 0x0FD3 which attestation covers a payment: `find_lowest_attested_after`, `get_attestation_bounds`.
- Proven against the live precompile before deploying: a real Sepolia proof → `0x0FD2.verify = true`; tampered bytes and a wrong chain key revert (pnpm `verify:live`).

ATTACK LAB

Eight ways to cheat: six decoded rejections, two accepted by design

- Replay → QueryAlreadyProcessed
- Fake vault with a perfect event → WrongEmitter
- Same proof, chain key 3 → WrongChain
- Included but reverted tx → SourceTxFailed
- Paid after the deadline → accepted, flagged late, −20
- Steal the steward’s key → NotOperator / OwnableUnauthorizedAccount / RoundStillOpenOnSource / VaultNotTrusted
- Fire the agent → accepted: the ledger checks the proof, not the caller
- Poison the reasoning → three invented figures stripped before display

Reviewed adversarially before submission: only owner-trusted vaults can feed the ledger, nobody can be penalised for a circle they never consented to, a 64-block grace window protects on-time payers from a hostile early close, and pots only go to members who paid this round.

KITTY STEWARD

An agent whose only power is proof

LAYER	HOLDS	WHAT IT DOES
1 · The ledger	final say	Verified proof, receipt status 1, trusted emitter, right sender and target, exact amount, current round, unseen query id. Any failure: nothing happens.
2 · Deterministic decisions	timing only	Prove now or wait for a fuller batch? Missing the grace window costs 120 points: urgency beats thrift. Decisions logged.
3 · Cited reasoning	none	Claude explains the log in plain language. Any sentence with an uncited or unverifiable figure is stripped before display.

No API key: the deterministic sentence, identical behaviour.

KITTY SCORE → CREDIT

The score is used, not just displayed

- 500 base · +15 on time · −20 late · −120 missed · 300–850.
- KittyCreditLine on Creditcoin underwrites from the score alone: tier A borrows 100% of proven volume, B 50%, C 20%, D nothing.
- A soulbound Kitty Score badge renders live from the ledger, so a lender can read it on any explorer.
- Every input is a proven transaction or an attested deadline. This is the data Creditcoin was built to carry.

LIVE ON TESTNET

The first circle has already settled

- KittyLedger, KittyViewer, KittyUSD, KittyCreditLine and KittyBadge deployed to Creditcoin CC3 Testnet on 12 September 2026 (KittyLedger `0xc6fe…c2dE`); the Sepolia vault `0x1172…284E` is trusted on the ledger.
- Delhi Chit Circle, three members, 100 tUSD a round, rotation by score: three Sepolia payments, two batch proofs verified by the live 0x0FD2, an early close, a 300 tUSD payout on Sepolia and its proof back to Creditcoin, all inside twenty minutes.
- Attack scenarios rerun against the real precompile: the forged chain key is rejected by 0x0FD2 itself before the ledger's own check can fire.
- Every transaction is linked in docs/TESTNET_LOG.md; the hosted dashboard, steward log and attack lab read the same chain.

ROADMAP · CEIP

From circles to credit lines

- Score-gated circle sizes and seat bidding.
- Lender integrations on Creditcoin.
- Attestcoin writability for payouts once audited; Ethereum mainnet chainKey on CC3 mainnet.
- Pot cover through proven-event insurance.

TEAM

Prashant · solo builder

Foundry · TypeScript · React. Repo: https://github.com/Prashant-thakur77/Kitty. Contracts on Sepolia and Creditcoin CC3 Testnet; 103 Foundry tests · 26 agent tests · 8 attack scenarios end to end in CI; verified against the live 0x0FD2.